..ARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | Full Marks : 100
Course: Microprocessor and ALP | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between microprocessor and microcontroller. Explain Bus organization of a microprocessor with block diagram. 7

   b) Define T-state. Draw the labeled timing diagram of the instruction ADD M. 8

2. a) Write an ALP to multiply two 8 bit numbers stored in the memory address D050H and E050H and if the result is less than 80H, save it to F050H else store it in FFFFH. 8

   b) How does 8085 subtract 34H from 12H? What are the changes seen in flags of 8085 processor? 7

3. a) Write a program in 8086 ALP to display "MICROPROCESSOR" in reverse order. 8

   OR

   Write an assembly language in 8086 which transfer 10 bytes of data stored in "array 1" to the new location "array 2".

   b) What is macro assembler? Differentiate between macros and procedures. 7

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4. a) What is the role of pipelining and segmentation in 8086 microprocessor? Explain. 7

   b) What is DMA? Draw and explain the architecture of 8237A DMA controller. 8

   OR

   Define parallel interface. Explain 8255 PPI.

5. a) How will you interface 4KB RAM and 2KB ROM with 8085 from location A000H without foldback memory? 7

   b) Draw and explain the architecture of 8251 USART along with its pins. 8

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6. a) Explain about IVT of 8086. 8

b) Explain the mechanism of handling multiple interrupts with proper diagram. 7

7. Write short notes on: **(Any two)** 2×5

a) Memory Segmentation

b) Synchronous Bus Vs Asynchronous Bus

c) Interrupt service procedure

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